6.1.1 INCREMENTED DATE

ALGORITHM

Step 1: Start

Step 2: Input day, month, year

Step 3: If year ≤ 0 → Print "Invalid Date" and Stop

Step 4: If month < 1 OR month > 12 → Print "Invalid Date" and Stop

Step 5: Determine number of days in the month:

If month = 1,3,5,7,8,10,12 → days = 31

If month = 4,6,9,11 → days = 30

If month = 2:

If leap year → days = 29

Else → days = 28

(Leap year condition:
year % 400 == 0 OR
(year % 4 == 0 AND year % 100 ≠ 0))

Step 6: If day < 1 OR day > days → Print "Invalid Date" and Stop

Step 7: If day < days

day = day + 1

Step 8: Else (day == days)

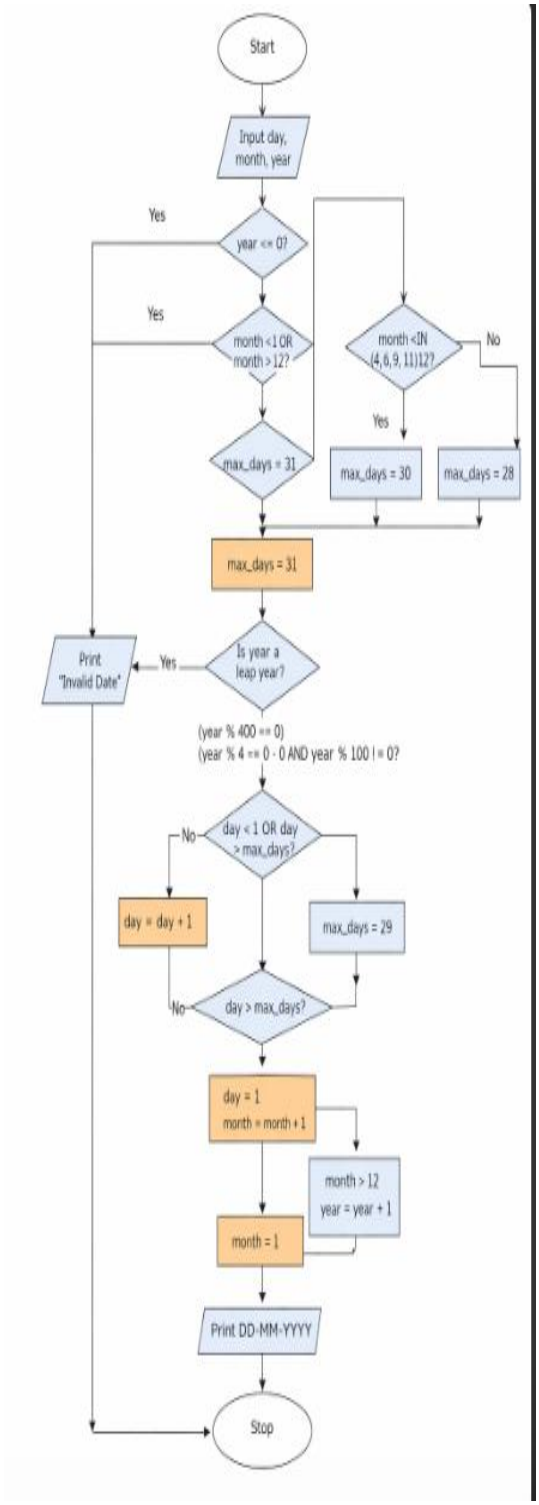
day = 1

If month == 12
month = 1
year = year + 1

Else
month = month + 1

Step 9: Print date in DD-MM-YYYY format

Step 10: Stop



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6.1.1. Incremented Date

01:52

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Sample Test Cases +

Explorer

nextDate.py

Submit

```python
day = int(input().strip())
month = int(input().strip())
year = int(input().strip())

def is_leap_year(y):
    return (y % 4 == 0 and y % 100 != 0) or (y % 400 == 0)

def days_in_month(m, y):
    if m in [1, 3, 5, 7, 8, 10, 12]:
        return 31
```

Debugger

Average time	Maximum time
0.004 s	0.009 s
4.50 ms	9.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output	Actual output
15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 2 3 ms

Terminal Test cases

‹ Prev Reset Submit Next ›